

결정학 및 회절(이,공) (MSE404a-01)

학기	2026년 2학기
담당교수	김진수
이수구분	교선평수
학점	3.0
강의시간	월10:30-12:00(E7 - G01), 수10:30-12:00(E7 - G01)
강의실	E6 315 (교수실)
소속	기초학부

교과개요

This course covers the fundamental principles of crystallography, which are essential for understanding the atomic arrangement and microstructure of materials. Students will learn key crystallographic concepts, including crystal lattices, symmetry, and crystal structures. Furthermore, the course will delve into the principles of X-ray diffraction (XRD), one of the most powerful tools for analyzing the crystal structure of materials. Students will explore in-depth methods for interpreting diffraction patterns to evaluate various material properties, such as phase analysis, lattice parameter calculation, and the determination of crystallite size and stress. This is an advanced course for 4th-year materials engineering students, providing a critical foundation for understanding the structure-property relationships in materials.

교과목표

Upon successful completion of this course, students will be able to: 1. Understand Basic Concepts of Crystallography: Explain concepts such as space lattice, unit cell, symmetry elements, point group, and space group. 2. Explain the Principles of X-ray Diffraction: Understand and describe the principles of X-ray generation, the interaction of X-rays with matter, Bragg's Law, and factors affecting diffraction intensity 3. Analyze and Interpret XRD Data: Identify crystal structures from given XRD patterns, precisely calculate lattice parameters, and perform qualitative and quantitative phase analysis. 4. Apply Knowledge to Evaluate Material Properties: Utilize peak broadening to calculate crystallite size and microstrain, and understand and apply the principles of residual stress measurement.

학습윤리

Unauthoritarian assistance is not allowed on all assignments and exams. Misrepresentation of your identity is not also permitted.

교과정책(방침)

The course will consist mainly of theoretical lectures, but will include some practical and case studies. Students are required to submit assignments on the given date, and late submissions will result in a deduction of points. Additional materials (papers, experimental results, etc.) will be provided when necessary. Midterm and final exams are not open-book.